

NLP PT1 Question Bank - Numerical Solutions

This document contains step-by-step explanations and math calculations for the numerical section (Questions Q1 to Q4) of the NLP PT1 Question Bank.

Q1. Viterbi Decoding for "the light book"

Given a Hidden Markov Model (HMM) for Part-of-Speech (POS) tagging:

- **Vocabulary:** $V = \{\text{"the"}, \text{"light"}, \text{"book"}\}$
- **Tags (Hidden States):** $S = \{DT, NN, VB, JJ\}$
- **Initial Probability vector (π):**

$\pi = [DT: 0.45, NN: 0.29, VB: 0.01, JJ: 0.25]$

Given Matrices

Transition Probability Matrix (A):

From \ To	DT	NN	VB	JJ
DT	0	0.5	0.00001	0.60
NN	0	0.2	0.30	0.0002
VB	0	0.3	0.10	0
JJ	0	0.2	0.001	0

Emission Probability Matrix (B):

Tag \ Word	the	light	book
DT	0.30	0	0
NN	0.10	0.003	0.003
VB	0	0.06	0.01
JJ	0	0.002	0

Step-by-Step Viterbi Execution

Let $V_t(s)$ represent the maximum probability of a state sequence ending in state s at time step t , emitting the first t words.

Step 1: Initialization ($t = 1$, Word $w_1 = \text{"the"}$)

$V_1(s) = \pi(s) \times B(s, \text{"the"})$

- $V_1(DT) = 0.45 \times 0.30 = 0.135$
 - $V_1(NN) = 0.29 \times 0.10 = 0.029$
 - $V_1(VB) = 0.01 \times 0 = 0$
 - $V_1(JJ) = 0.25 \times 0 = 0$
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Step 2: $t = 2$, Word $w_2 = \text{"light"}$

$$V_2(s_j) = \max_i [V_1(s_i) \times A(s_i, s_j)] \times B(s_j, \text{"light"})$$

1. **State $s_j = DT$:** Since $B(DT, \text{"light"}) = 0$, $V_2(DT) = 0$.

2. **State $s_j = NN$:**

- $B(NN, \text{"light"}) = 0.003$
- Candidate paths from $t=1$:
- From DT : $V_1(DT) \times A(DT, NN) = 0.135 \times 0.5 = 0.0675$ (Maximum)
- From NN : $V_1(NN) \times A(NN, NN) = 0.029 \times 0.2 = 0.0058$
- From VB : $0 \times A(VB, NN) = 0$
- From JJ : $0 \times A(JJ, NN) = 0$
- $V_2(NN) = 0.0675 \times 0.003 = 0.0002025$ (2.025×10^{-4})
- $\text{Backpointer } BP_2(NN) = DT$

3. **State $s_j = VB$:**

- $B(VB, \text{"light"}) = 0.06$
- Candidate paths from $t=1$:
- From DT : $0.135 \times 0.00001 = 1.35 \times 10^{-6}$
- From NN : $0.029 \times 0.30 = 0.0087$ (Maximum)
- From others: 0
- $V_2(VB) = 0.0087 \times 0.06 = 0.000522$ (5.22×10^{-4})
- $\text{Backpointer } BP_2(VB) = NN$

4. **State $s_j = JJ$:**

- $B(JJ, \text{"light"}) = 0.002$
 - Candidate paths from $t=1$:
 - From DT : $0.135 \times 0.60 = 0.081$ (Maximum)
 - From NN : $0.029 \times 0.0002 = 5.8 \times 10^{-6}$
 - From others: 0
 - $V_2(JJ) = 0.081 \times 0.002 = 0.000162$ (1.62×10^{-4})
 - $\text{Backpointer } BP_2(JJ) = DT$
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Step 3: $t = 3$, Word $w_3 = \text{"book"}$

$$V_3(s_j) = \max_i [V_2(s_i) \times A(s_i, s_j)] \times B(s_j, \text{"book"})$$

1. **State $s_j = DT$:** Since $B(DT, \text{"book"}) = 0$, $V_3(DT) = 0$.

2. **State $s_j = NN$:**

- $B(NN, \text{"book"}) = 0.003$
- Candidate paths from $t=2$:
- From DT : $0 \times 0.5 = 0$
- From NN : $2.025 \times 10^{-4} \times 0.2 = 4.05 \times 10^{-5}$
- From VB : $5.22 \times 10^{-4} \times 0.3 = 1.566 \times 10^{-4}$ (Maximum)
- From JJ : $1.62 \times 10^{-4} \times 0.2 = 3.24 \times 10^{-5}$
- $V_3(NN) = 1.566 \times 10^{-4} \times 0.003 = 4.698 \times 10^{-7}$
- $\text{Backpointer } BP_3(NN) = VB$

3. **State $s_j = VB$:**

- $B(VB, \text{"book"}) = 0.01$
- Candidate paths from $t=2$:
- From DT : 0
- From NN : $2.025 \times 10^{-4} \times 0.3 = 6.075 \times 10^{-5}$ (Maximum)
- From VB : $5.22 \times 10^{-4} \times 0.1 = 5.22 \times 10^{-5}$
- From JJ : $1.62 \times 10^{-4} \times 0.001 = 1.62 \times 10^{-7}$
- $V_3(VB) = 6.075 \times 10^{-5} \times 0.01 = 6.075 \times 10^{-7}$
- $\text{Backpointer } BP_3(VB) = NN$

4. **State $s_j = JJ$:** Since $B(JJ, \text{"book"}) = 0$, $V_3(JJ) = 0$.

Backtracking & Final Sequence

- At $t = 3$, the state with the maximum probability is:

$\max(V_3(DT), V_3(NN), V_3(VB), V_3(JJ)) = \max(0, 4.698 \times 10^{-7}, 6.075 \times 10^{-7}, 0) = 6.075 \times 10^{-7}$ This occurs at state **VB**.

- **Traceback:**
- $t = 3$: **VB**
- $t = 2$: $BP_3(VB) = NN$
- $t = 1$: $BP_2(NN) = DT$

The most likely tag sequence is: **DT** → **NN** → **VB** (Probability: 6.075×10^{-7}).

> [!NOTE] > **Comparison with DT JJ NN:** $P(\text{"the/DT light/JJ book/NN"}) = \pi(DT) \times B(DT, \text{"the"}) \times A(DT, JJ) \times B(JJ, \text{"light"}) \times A(JJ, NN) \times B(NN, \text{"book"}) > P = 0.45 \times 0.30 \times 0.60 \times 0.002 \times 0.2 \times 0.003 = 9.72 \times 10^{-8}$ > Since $6.075 \times 10^{-7} > 9.72 \times 10^{-8}$, **DT NN VB** is indeed the globally optimal path.

Q2. Find the Likelihood of "Book/VB Ticket/NN"

Given the Tag sequence $T = (VB, NN)$ for the word sequence $W = (\text{"Book"}, \text{"Ticket"})$.

Given Parameters

Transition Probabilities (A):

- $P(NN | <s>) = 0.7$, $P(VB | <s>) = 0.3$
- $P(NN | VB) = 0.7$, $P(VB | VB) = 0.2$

Emission Probabilities (B):

- $P(\text{"Book"} | VB) = 0.3$, $P(\text{"Ticket"} | VB) = 0.7$
- $P(\text{"Book"} | NN) = 0.4$, $P(\text{"Ticket"} | NN) = 0.6$

Joint Likelihood Calculation

The joint likelihood of the word-tag sequence is: $P(W, T) = P(VB | <s>) \times P(\text{"Book"} | VB) \times P(NN | VB) \times P(\text{"Ticket"} | NN)$

Substitute the parameter values: $P(W, T) = 0.3 \times 0.3 \times 0.7 \times 0.6$ $P(W, T) = 0.09 \times 0.42 = 0.0378$ (3.78%)

Q3. HMM POS Tagger from Corpus

Given Corpus

- $<s> \text{ a/DT dog/NN chases/V a/DT cat/NN } </s>$
- $<s> \text{ the/DT dog/NN barks/V loudly/RB } </s>$
- $<s> \text{ a/DT cat/NN runs/V fast/RB } </s>$

i) Transition and Emission Probabilities

1. Transition Counts and Probabilities

From State	To State	Count	Transition Probability $P(T_j T_i)$
$<s>$	DT	3	$3/3 = 1.0$
DT	NN	4	$4/4 = 1.0$
NN	V	3	$3/4 = 0.75$
	$</s>$	1	$1/4 = 0.25$
V	DT	1	$1/3 \approx 0.333$
	RB	2	$2/3 \approx 0.667$
RB	$</s>$	2	$2/2 = 1.0$

2. Emission Counts and Probabilities

- **DT** (Total Count = 4):
 - **a**: 3 $\implies P(\text{"a"} | DT) = 3/4 = 0.75$
 - **the**: 1 $\implies P(\text{"the"} | DT) = 1/4 = 0.25$
 - **NN** (Total Count = 4):
 - **dog**: 2 $\implies P(\text{"dog"} | NN) = 2/4 = 0.50$
 - **cat**: 2 $\implies P(\text{"cat"} | NN) = 2/4 = 0.50$
 - **V** (Total Count = 3):
 - **chases**: 1 $\implies P(\text{"chases"} | V) = 1/3 \approx 0.333$
 - **barks**: 1 $\implies P(\text{"barks"} | V) = 1/3 \approx 0.333$
 - **runs**: 1 $\implies P(\text{"runs"} | V) = 1/3 \approx 0.333$
 - **RB** (Total Count = 2):
 - **loudly**: 1 $\implies P(\text{"loudly"} | RB) = 1/2 = 0.50$
 - **fast**: 1 $\implies P(\text{"fast"} | RB) = 1/2 = 0.50$
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ii) Decode: "The cat chases the dog"

- **Words:** $w_1 = \text{"the"}, w_2 = \text{"cat"}, w_3 = \text{"chases"}, w_4 = \text{"the"}, w_5 = \text{"dog"}$.

Step 1: $t=1$, "the"

- $V_1(DT) = P(DT | <s>) \times P(\text{"the"} | DT) = 1.0 \times 0.25 = 0.25$
- Others are 0.

Step 2: $t=2$, "cat"

- $V_2(NN) = V_1(DT) \times P(NN | DT) \times P(\text{"cat"} | NN) = 0.25 \times 1.0 \times 0.5 = 0.125$ ($BP_2 = DT$)
- Others are 0.

Step 3: $t=3$, "chases"

- $V_3(V) = V_2(NN) \times P(V | NN) \times P(\text{"chases"} | V) = 0.125 \times 0.75 \times (1/3) = 0.03125$ ($BP_3 = NN$)
- Others are 0.

Step 4: $t=4$, "the"

- $V_4(DT) = V_3(V) \times P(DT | V) \times P(\text{"the"} | DT) = 0.03125 \times (1/3) \times 0.25 = 0.002604$ ($BP_4 = V$)
- Others are 0.

Step 5: $t=5$, "dog"

- $V_5(NN) = V_4(DT) \times P(NN | DT) \times P(\text{"dog"} | NN) = 0.002604 \times 1.0 \times 0.5 = 0.001302$ ($BP_5 = DT$)

Backtracking:

- $t=5$: **NN**
- $t=4$: $BP_5(NN) = DT$
- $t=3$: $BP_4(DT) = V$
- $t=2$: $BP_3(V) = NN$

- $t=1$: $BP_2(NN) = \mathbf{DT}$

Final Sequence: $\mathbf{DT} \rightarrow \mathbf{NN} \rightarrow \mathbf{V} \rightarrow \mathbf{DT} \rightarrow \mathbf{NN}$

Q4. Bigram Language Model

Unigram Counts (Case-Insensitive)

Total sentences $N = 7$.

- **<s>**: 7
 - **I**: 6
 - **am**: 2
 - **Sam**: 5
 - **like**: 5
 - **college**: 3
 - **do** (including **Do**): 4
 - **</s>**: 7
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Bigram Counts

- **<s>** followed by: **I** (3), **do/Do** (3), **Sam** (1)
 - **I** followed by: **am** (2), **like** (3), **do** (1)
 - **am** followed by: **Sam** (1), **</s>** (1)
 - **Sam** followed by: **</s>** (3), **like** (1), **I** (1)
 - **like** followed by: **college** (3), **Sam** (2)
 - **college** followed by: **</s>** (3)
 - **do/Do** followed by: **I** (2), **like** (1), **Sam** (1)
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Predict Next Words

i) Next word after "like"

- $P(\text{college} \mid \text{like}) = \frac{C(\text{like, college})}{C(\text{like})} = \frac{3}{5} = 0.60$
- $P(\text{Sam} \mid \text{like}) = \frac{C(\text{like, Sam})}{C(\text{like})} = \frac{2}{5} = 0.40$

Prediction: college (since $0.60 > 0.40$).

ii) Next word after "Sam"

- $P(\text{</s>} \mid \text{Sam}) = \frac{C(\text{Sam, </s>})}{C(\text{Sam})} = \frac{3}{5} = 0.60$
- $P(\text{like} \mid \text{Sam}) = \frac{C(\text{Sam, like})}{C(\text{Sam})} = \frac{1}{5} = 0.20$
- $P(\text{I} \mid \text{Sam}) = \frac{C(\text{Sam, I})}{C(\text{Sam})} = \frac{1}{5} = 0.20$

Prediction: **</s>** (Sentence boundary, since 0.60 is highest).

Sentence Comparison: Which is better?

Sentence 1 (S₁): **<s> Sam I am </s>**

$$P(S_1) = P(\text{Sam} | <s>) \times P(\text{I} | \text{Sam}) \times P(\text{am} | \text{I}) \times P(</s> | \text{am}) \\ P(S_1) = \frac{1}{7} \times \frac{1}{5} \times \frac{2}{6} \times \frac{1}{2} = \frac{1}{7} \times \frac{1}{5} \times \frac{1}{3} \times \frac{1}{2} = \frac{1}{210} \approx 0.00476$$

Sentence 2 (S₂): **<s> I do like Sam </s>**

- **Case 1: Case-Insensitive Model** (treating **Do** and **do** as same token **do**):

$$P(S_2) = P(\text{I} | <s>) \times P(\text{do} | \text{I}) \times P(\text{like} | \text{do}) \times P(\text{Sam} | \text{like}) \times P(</s> | \text{Sam}) \\ P(S_2) = \frac{3}{7} \times \frac{1}{6} \times \frac{1}{4} \times \frac{2}{5} \times \frac{3}{5} = \frac{18}{4200} \approx 0.00428$$

*Under Case-Insensitive assumptions, $P(S_1) \approx 0.00476 > P(S_2) \approx 0.00428$, so **Sentence 1** is slightly better.*

- **Case 2: Case-Sensitive Model** (treating capitalized **Do** and lowercase **do** separately):

$$\text{Since } \text{do} \text{ only appears once (in } S_7 \text{) and is followed by } \text{like: } P(\text{like} | \text{do}) = \frac{1}{1} = 1.0 \\ P(S_2) = \frac{3}{7} \times \frac{1}{6} \times 1.0 \times \frac{2}{5} \times \frac{3}{5} = \frac{18}{1050} \approx 0.0171$$

*Under Case-Sensitive assumptions, $P(S_2) \approx 0.0171 > P(S_1) \approx 0.00476$, making **Sentence 2** significantly better.*